Vaibhav Lalwani

AI Systems Engineer | Applied AI and forward-deployed engineering | MSc Advanced Data Science & AI, University of Liverpool

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Open to UK part-time roles and internships in applied AI, ML engineering and forward-deployed engineering | Graduate roles from January 2027

PROFILE

AI systems engineer building production LLM applications, retrieval pipelines and agent products end to end, then fitting them into the customer's own workflow: scoping, evals before rollout, deployment and hand-off. Four hackathon wins in 2026; 20+ AI applications shipped, with 20+ clients served one-on-one in their offices.

SKILLS

AI systems	Claude, OpenAI, Gemini and ADK, Groq, NVIDIA NIM, Ollama, AssemblyAI speech, QLoRA; MCP and WebMCP, multi-agent orchestration, structured outputs, approval gates; RAG, hybrid search and rank fusion, reranking, citations; eval sets, regression gates, LangSmith
Languages	TypeScript, Python, Rust (STILL core service, ATKIN's Tauri backend), Go (Dolphin Tank), SQL, Solidity
Product	React 19, Next.js 15, Node.js, FastAPI, Tauri, Three.js, WebSockets, Web Push, REST APIs, Vitest, Playwright
Data and ship	Postgres and pgvector, MongoDB Atlas Vector Search, Supabase, Redis and BullMQ, HydraDB, Docker, Vercel, GCP Cloud Run, GitHub Actions, Sentry, PostHog
Delivery	Scoping with the client team, existing-workflow fit, eval set agreed before rollout, runbook and video hand-off, privacy by design, authorised security testing

EXPERIENCE

Instructor | Robocode UK — Aug 2026 - Present
Part-time, Liverpool, on-site

- Teaching coding and robotics to young learners in Liverpool.

AI Engineer, Full-Stack | Meta Solution Technologies — Apr - May 2026
Part-time, remote

- AI-guided admissions platform for international students applying to UK and US universities: Next.js 15, React 19 and TypeScript over a Python LLM backend; Zod-validated structured outputs, prompts regressed against a fixed eval set before each rollout; RAG on Postgres and pgvector with reranking and citations; BullMQ and Redis, Docker, Cloud Run, LangSmith, Sentry, PostHog.

Software Engineer | RecruitPilot AI — Jan - Apr 2026
Part-time, remote

- Built React and TypeScript features against a REST API for a recruitment-tech product, in code review and a weekly sprint cadence alongside senior engineers.

AI Engineer, Freelance | Self-employed — May 2024 - Jan 2026
Remote and on-site

- Delivered 20+ custom AI applications for SaaS, e-commerce, real estate and consulting clients, and built and sold 5 proprietary AI agents.
- Worked one-on-one with 20+ clients in their offices, on their workflows; every engagement ended with a README, a walkthrough video and a hand-off document.

Front-End Developer, Internship | Unified Mentor — Mar - May 2025
Remote

- Built React UI features on a learner-facing education platform: lessons, progression and learner dashboards against a REST API.

SELECTED WORK

Yuzu - SharedOS Hackathon Judges' Pick. A market where agents bid, prove capability, contract, execute and settle with signed receipts. Source

Leverage - RocketRide x SCU Buildathon winner. An auction hires cheaper models per task; the repository's own tests decide what counts. Source

HydraSentry - HydraDB Build Blitz winner. Agent-memory integrity layer that blocks unsafe context and signs a review certificate. Source

QueueProof - HydraDB Connectors Hackathon winner. Cross-source retrieval with claim-level citations back to the source record. Source

Stay - Voice companion for adults with ADHD on AssemblyAI streaming speech. Submitted to AssemblyAI Voice Hackathon Week. Source

Qyntra - Wikithon 2026 finalist. Personal knowledge system with grounded answers and a 3D relationship map. Source

STILL and ATKIN - Rust systems work: STILL's core service for GitLab Duo repairs, ATKIN's Tauri backend for a local-first legal workspace. Source

SECURITY, RESEARCH AND EDUCATION

Hack HydraDB (2026): top 3 of 20+ in an authorised security round, and 6 defects filed in a second authorised read-only bug bash; coordinated disclosure, specifics withheld. **NEXUS** (2026): corrected v2 position paper on on-controller transformer inference; architecture and analytical model, no prototype results. Corrected v2

University of Liverpool — Jan 2026 - Jan 2027
MSc Advanced Data Science & Artificial Intelligence

Christ University, Bengaluru — 2022 - 2025
Bachelor of Computer Applications